AI SYSTEMS R&D

Xander Xu

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AI systems engineer and Core-AI open-source contributor, working on durable agent execution, recovery consistency and model-protocol integration. Earlier work spans biomedical machine learning; recent competition projects cover LLM inference and policy deployment.

Technical capabilities

Runtime	Java, Python; agent workflows, pause/resume, state persistence, concurrency
Interfaces & state	OpenAI-compatible / Responses APIs, MCP; MongoDB, Redis, execution tracing
Inference	vLLM, NumPy; tool-augmented inference, sampling, early stopping, policy deployment

Recent experience

Jul 15, 2025 – Present
China

ChanceTop · AI Engineer

- Implemented human-input pause/resume and persisted workflow state in the team-built Core-AI framework; serialized snapshot-and-write operations to prevent stale overwrites during cancellation, with a concurrency regression test.
- Added Redis heartbeat/TTL-based cross-pod execution-state detection and an unknown-state fallback, fixing erroneous clearing of active chat messages; added recovery and failure-path regression coverage.
- Implemented capability-aware Responses routing for reasoning models with tools, preserving other model routes and adding compatibility regression cases.
- Applied agent workflows to structured data processing and Local SEO automation.

Sep 2024 – Jul 2025
Xiamen, China

Qudian Technology · LLM Application Engineer

- Designed multimodal agent workflows combining retrieval, function calling, image generation and speech interfaces, with prompt and content-safety controls.

Mar 2023 – Jun 2024
Xiamen, China

Jiyuan Technology · Technical Lead, Medical LLM R&D

- Integrated an oncology knowledge base through RAG, applied LoRA for domain adaptation, and developed intent-recognition and entity-extraction workflows.

Selected technical projects

AIMO3 · Tool-augmented mathematical reasoning

2026 · Kaggle bronze · 405 / 4,138 teams

- Configured gpt-oss-120B/vLLM on a public tool-integrated reasoning baseline with eight parallel attempts, persistent Python execution, four-vote early stopping and a dynamic time budget.
- Modified answer aggregation to weight Python-call counts and error rates as confidence heuristics; published the inspectable inference notebook.

Orbit Wars · Self-play policy experiments

2026 · Kaggle silver · 112 / 4,729 teams

- Worked on an existing BC→PPO self-play stack, adding reward experiments, JAX runtime configuration and persistent compilation caching.
- Integrated separate 2P/4P policies into a router with a NumPy deployment path requiring no PyTorch; used official-environment and paired-seed evaluation gates for candidate acceptance.

Earlier algorithm & research experience

Apr 2022 — Mar 2023
Shenzhen, China

Fapon Biotech · Algorithm Engineer and Project Lead

- Led algorithm design for recurrence-risk stratification from whole-slide pathology images.

Apr 2021 — Apr 2022
Shenzhen, China

BGI Research · Algorithm Engineer and Team Lead

- Developed attention-enhanced ssDNA image segmentation and proposed GMM-based Stereo-seq correction using spatial and expression information.

Oct 2018 — Apr 2021
Xiamen, China

Jiyuan Technology · Algorithm Engineer

- Developed pathology-image pipelines with U-Net and convolutional features; co-first author of studies on metagenomic geolocation and anti-PD-1 response.

Jul 2017 — Oct 2018
Beijing, China

Tsinghua University, Institute for Interdisciplinary Information Sciences · Research Assistant

- Built heterogeneous-network features for drug–target ranking and applied deep-learning detection to low-signal cryo-EM images.

Selected research

1. **Ratio of the interferon-γ signature to the immunosuppression signature predicts anti-PD-1 therapy response in melanoma**
   npj Genomic Medicine 6, 7 (2021) · Co-first author
2. **A machine learning framework to determine geolocations from metagenomic profiling**
   Biology Direct 15, 27 (2020) · Co-first author
3. **Inconsistent prediction capability of ImmuneCells.Sig across different RNA-seq datasets**
   Nature Communications 12, 4167 (2021) · Co-author

Competitions & technical communication

Kaggle Notebooks Master · 6 gold · 4 silver notebook medals
Competitions Expert, with a further 2 silver and 5 bronze competition medals across modeling and agent challenges.

过拟合青年 · Bilibili · 24,091 followers · Animated explanations of AI mechanisms

Updated 29 Oct 2026

Education

Fujian Medical University · BSc in Bioinformatics · 2013 — 2017